Lightweight Anomaly Detection for Real-Time Network Traffic in Edge/IoT Environments

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***Abstract*—Intrusion Detection Systems (IDS) play a crucial role in securing modern networks; however, most existing solutions are computationally intensive and unsuitable for deployment in resource-constrained edge and IoT environments. This work focuses on a lightweight anomaly detection framework using Isolation Forest, optimized for real-time network traffic analysis. The system is trained exclusively on benign traffic to model normal behavior and detect anomalies effectively. In addition, a deep learning-based Autoencoder is implemented for comparative analysis. Experiments conducted on the IoT-23 and CICIDS2017 datasets demonstrate that the proposed approach achieves a strong balance between detection accuracy and computational efficiency. On IoT-23, the final model achieves an attack recall of 87.7%, precision of 98.9%, false positive rate of approximately 10%, and an AUC score of 0.93. On CICIDS2017, the model achieves an attack recall of 63.4%, precision of 89.2%, false positive rate of approximately 10%, and an AUC score of 0.81, while maintaining microsecond-level latency.**

***Index Terms*—Intrusion Detection System, IoT Security, Edge Computing, Anomaly Detection, Isolation Forest, Autoencoder, CICIDS2017**

I. Introduction

The rapid proliferation of Internet of Things (IoT) devices has significantly increased the attack surface of modern networks. These devices, including smart cameras, routers, and home automation systems, operate under strict resource constraints such as limited processing power and memory.

Traditional intrusion detection systems are designed for high-performance environments and are not suitable for deployment on edge devices. Furthermore, signature-based methods fail to detect unknown attacks, while many machine learning approaches are computationally expensive.

This work focuses on designing a lightweight, real-time anomaly detection system that can operate efficiently in edge environments while maintaining high detection accuracy.

II. Related Work

Several studies have explored anomaly detection techniques for network security. Isolation Forest has been widely used due to its efficiency and suitability for high-dimensional data. Liu et al. introduced Isolation Forest as a scalable method for anomaly detection based on random partitioning.

Deep learning approaches, such as Autoencoders, have also been applied to intrusion detection tasks. These models learn latent representations of normal data and detect anomalies based on reconstruction error. However, their effectiveness depends heavily on the complexity of the data and computational resources.

Recent research highlights that while deep learning models can capture complex patterns, simpler machine learning models often outperform them in structured, tabular datasets commonly found in network traffic analysis.

III. Proposed Methodology

A. System Architecture

The proposed framework processes network flow data through feature extraction, preprocessing, model training, and anomaly detection. Fig. 1 illustrates the overall system architecture.

B. Datasets

Two datasets were used in this study to evaluate both performance and generalization capability.

IoT-23 Dataset: The IoT-23 dataset contains labeled network traffic from IoT devices, including both benign and malicious activities. It provides Zeek-generated connection logs (`conn.log.labeled`) with flow-level features.

CICIDS2017 Dataset: CICIDS2017 was used as a second dataset to evaluate the robustness of the proposed approach on more complex and diverse network traffic. This dataset contains benign traffic and multiple attack types, including DDoS traffic, making it suitable for validating generalization performance.

C. Data Preprocessing

The preprocessing pipeline includes:

- Removal of metadata rows where applicable
- Replacement of missing and infinite values with numerical values
- Selection of relevant flow-based features
- Encoding of categorical features where required
- Feature normalization using StandardScaler
- Binary label conversion (Benign = 0, Malicious = 1)

For CICIDS2017, selected lightweight features included flow duration, packet counts, packet lengths, flow bytes per second, flow packets per second, packet length mean, packet length standard deviation, and average packet size.

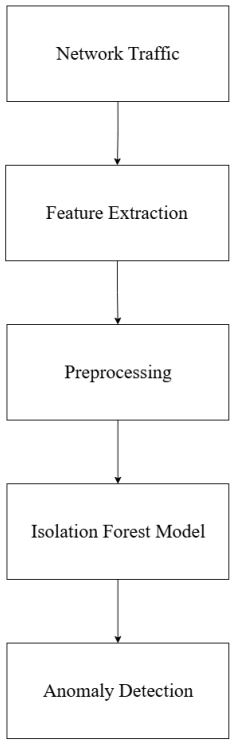

Fig. 1. Proposed System Architecture

D. Isolation Forest Model

Isolation Forest is an unsupervised anomaly detection algorithm that isolates anomalies by randomly partitioning data. Normal data points typically require more splits to isolate, while anomalous points are isolated quickly.

The anomaly score can be expressed as:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \tag{1}$$

where $E(h(x))$ is the expected path length of a sample x, and $c(n)$ is the average path length of unsuccessful searches in a binary search tree. A higher anomaly score indicates a higher likelihood of being anomalous.

The model was trained exclusively on benign traffic to capture normal behavior patterns.

E. Autoencoder Model

An Autoencoder was implemented as a deep learning comparison. It consists of an encoder and decoder network that reconstructs input data. Normal data is expected to produce low reconstruction error, while anomalous data produces higher reconstruction error.

The reconstruction loss is calculated using Mean Squared Error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \tag{2}$$

where x_i is the original input and $\hat{x}_i$ is the reconstructed output.

IV. Experimental Setup

Experiments were conducted using Python with Scikit-learn and TensorFlow/Keras.

- Training on benign-only data
- Contamination parameter tuning
- Feature reduction for efficiency
- Evaluation using precision, recall, FPR, latency, and AUC
- Cross-dataset validation using IoT-23 and CICIDS2017

The false positive rate was calculated as:

$$FPR = \frac{FP}{FP + TN} \tag{3}$$

where FP represents false positives and TN represents true negatives.

V. Results and Analysis

A. Baseline Model

Initial training on mixed data resulted in poor performance due to model confusion between normal and anomalous patterns.

B. Improved Model

Training only on normal data significantly improved detection performance.

C. Contamination Tuning

The contamination parameter was tuned to balance detection and false positives. The optimal value was:

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contamination = 0.1
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D. Feature Optimization

TABLE I
Feature Optimization Results on IoT-23

Features	Recall	FPR	Latency
12	99.6%	∼10%	8 μs
6	20%	∼10%	5 μs
8	87.7%	∼10%	5.7 μs

The results in Table I show that aggressive feature reduction reduces detection performance, while the balanced 8-feature model provides a better trade-off between accuracy and efficiency.

E. Model Comparison

The ROC curve in Fig. 5 shows the trade-off between the true positive rate and false positive rate. The proposed Isolation Forest model achieved an AUC of 0.93 on IoT-23, indicating strong capability in distinguishing between benign and malicious network traffic.

TABLE II
ISOLATION FOREST AND AUTOENCODER COMPARISON ON IOT-23

Model	Recall	FPR	AUC	Latency
Isolation Forest	87.7%	∼10%	0.93	∼4.7 μs
Autoencoder	<10%	∼3%	–	Higher

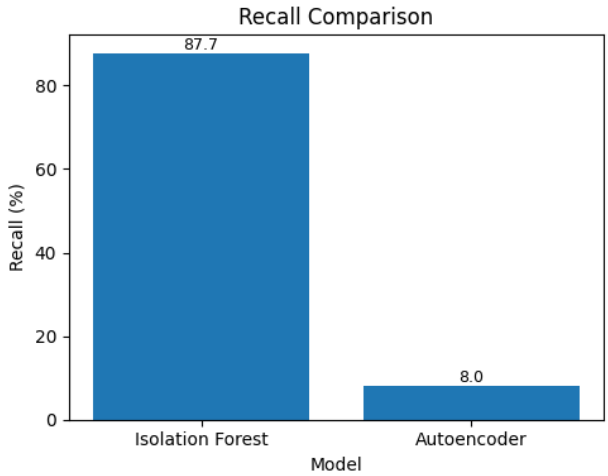

Fig. 2. Recall Comparison of Models

F. Evaluation on CICIDS2017 Dataset

To assess the generalization capability of the proposed model, experiments were also conducted on the CICIDS2017 dataset. The model achieved an AUC score of 0.81, attack recall of 63.4%, precision of 89.2%, and a false positive rate of approximately 10%. The latency per flow was approximately 3.7 μs.

TABLE III
CROSS-DATASET PERFORMANCE COMPARISON

Dataset	Recall	Precision	FPR	AUC
IoT-23	87.7%	98.9%	∼10%	0.93
CICIDS2017	63.4%	89.2%	∼10%	0.81

The results in Table III indicate that while the proposed model performs strongly on the IoT-23 dataset, its performance decreases on the more complex CICIDS2017 dataset. This demonstrates the difficulty of generalizing lightweight anomaly detection models across diverse network environments. However, the model maintains a stable false positive rate and high precision, supporting its suitability for practical real-time deployment.

G. Comparison with Supervised Models

Although supervised models such as Random Forest and Support Vector Machine (SVM) achieve higher accuracy and recall, they require fully labeled datasets and involve higher computational cost. In contrast, the proposed Isolation Forest model operates in an unsupervised manner and maintains

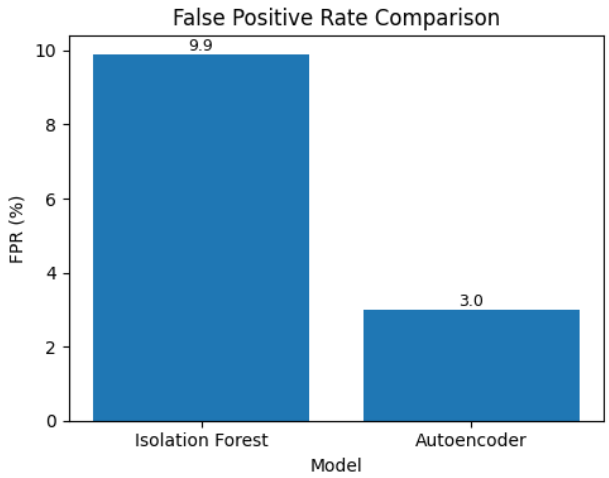

Fig. 3. False Positive Rate Comparison of Models

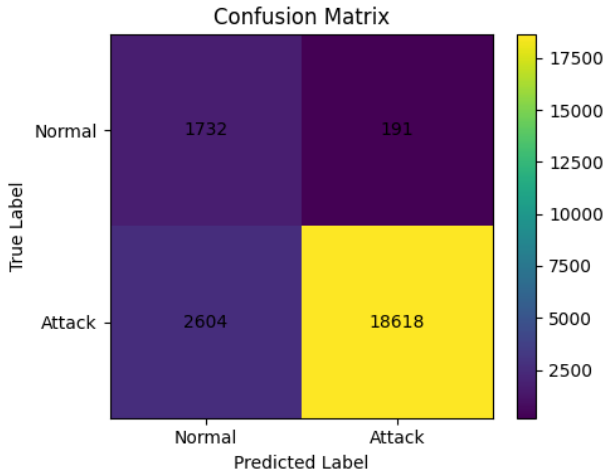

Fig. 4. Confusion Matrix of the Isolation Forest Model on IoT-23

significantly lower latency, making it more suitable for real-time deployment in resource-constrained edge and IoT environments.

H. Key Findings

- Isolation Forest outperforms Autoencoder for tabular network data.
- Deep learning models require more complex data patterns to perform effectively.
- Feature reduction must be balanced to maintain detection performance.
- The final model maintains low latency, making it suitable for edge deployment.
- Cross-dataset evaluation shows that the model generalizes reasonably but performs better on IoT-specific traffic than on more diverse network traffic.

VI. CONTRIBUTIONS

This work makes the following contributions:

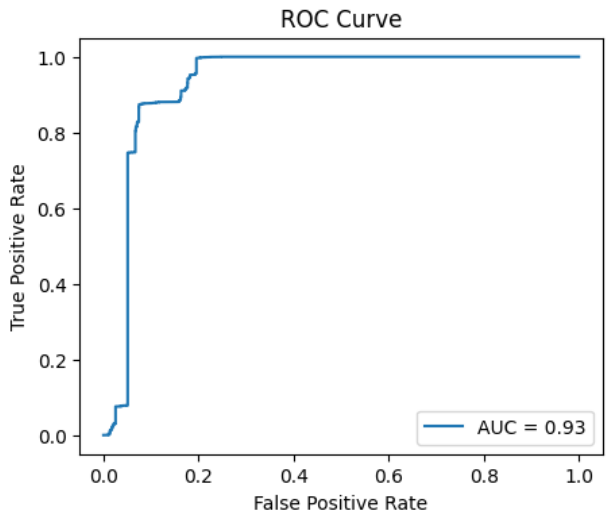

Fig. 5. ROC Curve of the Isolation Forest Model on IoT-23

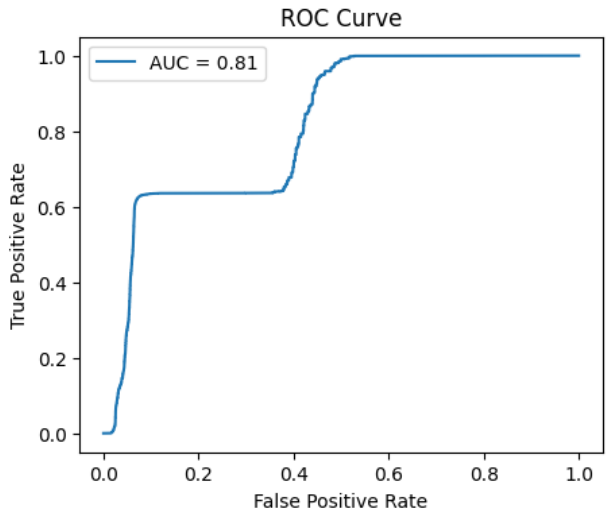

Fig. 6. ROC Curve of the Isolation Forest Model on CICIDS2017

- Development of a lightweight anomaly detection system for IoT and edge environments.
- Optimization of Isolation Forest through feature reduction and parameter tuning.
- Comparative analysis between machine learning and deep learning approaches.
- Cross-dataset validation using IoT-23 and CICIDS2017 datasets.
- Demonstration of the trade-off between detection performance and computational efficiency.

VII. Discussion

The results highlight an important insight: model selection should depend on data characteristics rather than model complexity. Isolation Forest provides an efficient and scalable solution for anomaly detection in structured datasets, while deep learning approaches may not always be suitable.The CICIDS2017 dataset exhibits class imbalance, which impacts detection performance, particularly recall. This reflects real-world network scenarios where attack traffic may be less frequent or highly variable. As a result, achieving high recall while maintaining low false positive rates remains a challenging task for lightweight anomaly detection models.

TABLE IV
Comparison with Random Forest and SVM

Model	Recall	Precision	FPR	Training Time
Isolation Forest	87.7%	98.9%	∼10%	Low
Random Forest	100%	100%	0%	Medium
SVM	99.2%	99.9%	0.3%	High

Although the Autoencoder achieved a lower false positive rate, it failed to detect most attack traffic. This indicates that low false positives alone are not sufficient for practical IDS deployment; high recall is also essential to minimize missed attacks.

The evaluation across multiple datasets highlights an important trade-off. While lightweight models offer efficiency and real-time capability, their performance may decrease in more complex environments. This reinforces the need to balance detection accuracy with computational constraints when designing practical intrusion detection systems for edge and IoT environments.

A. Limitations

Despite strong performance, the proposed model shows reduced recall on more complex datasets such as CICIDS2017. This indicates that lightweight models may struggle to capture highly diverse attack patterns. Additionally, the model relies on fixed anomaly thresholds, which may not adapt well to dynamically changing network environments.

VIII. Conclusion

This paper presented a lightweight anomaly detection system for network traffic analysis using Isolation Forest. The proposed approach achieved strong performance on the IoT-23 dataset and maintained reasonable performance on the more complex CICIDS2017 dataset. The results demonstrate that while performance decreases in more challenging environments, the model remains efficient, lightweight, and suitable for real-time edge deployment.

The comparative analysis revealed that the Autoencoder model underperformed due to the tabular nature of the dataset, reinforcing the effectiveness of simpler machine learning approaches for structured network data. Overall, the study highlights the importance of balancing accuracy and efficiency in practical IDS design.

IX. Future Work

Future improvements include:

- Deployment on real IoT hardware
- Sliding window-based real-time detection

- Hybrid models combining machine learning and deep learning
- Evaluation on additional datasets and attack scenarios
- Adaptive threshold tuning for improving recall on complex datasets

REFERENCES

[1] S. Garcia, A. Parmisano, and M. J. Erquiaga, "IoT-23: A labeled dataset with malicious and benign IoT network traffic," 2020.

[2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation Forest," in *Proceedings of the IEEE International Conference on Data Mining*, 2008.

[3] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proceedings of the International Conference on Information Systems Security and Privacy*, 2018.